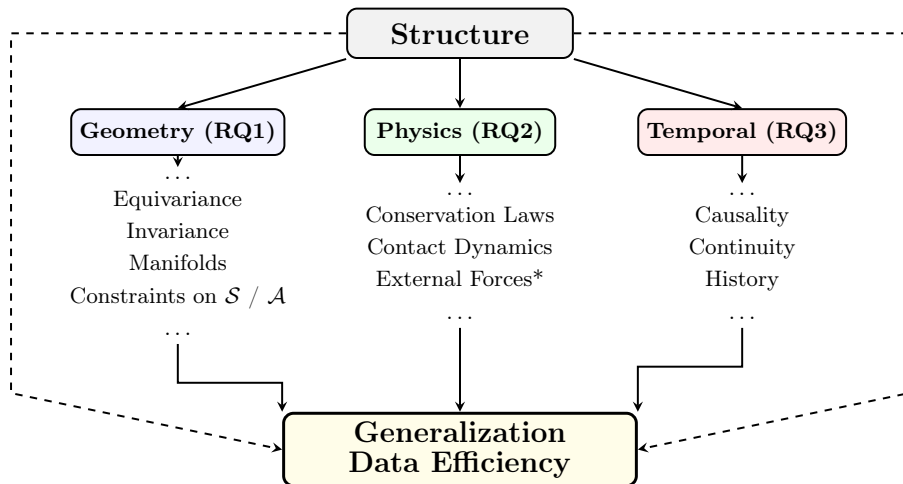
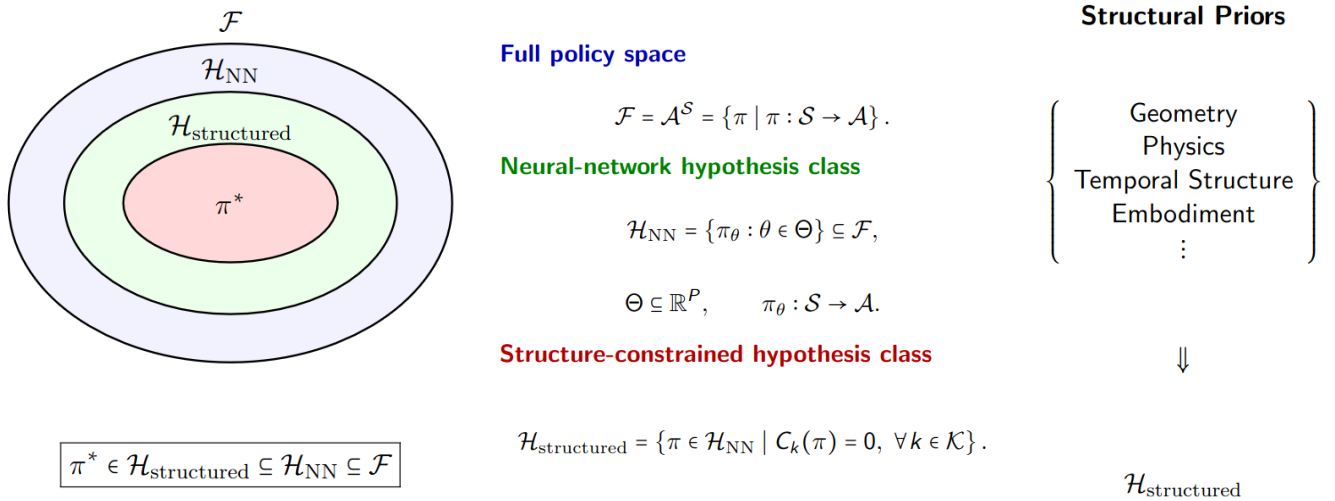


**Research Vision:** My research interest lies in the **intersection of robotics, machine learning, and control**, with a focus on developing physical intelligence through structural inductive biases. I am particularly interested in how control policies can leverage structures through **geometric and physics-informed priors** to constrain their hypothesis space and generalise under distributional shifts, such as changes in object geometry, contact dynamics, or embodiment.

Learning-based agents often fail to exploit such *a priori* structure, resulting in poor sample efficiency and limited generalisation. Without appropriate inductive biases, policy learning operates over a weakly constrained policy function class and may overfit to environment-specific features in the training distribution rather than capturing invariant geometric and physical relationships, limiting their ability to generalise in real-world settings.

To address this bottleneck, I plan to investigate techniques that **encode structural inductive biases and capture task invariants** [1] to improve data efficiency and generalisation capabilities in real-world robotic control. My ultimate goal is to contribute broadly to the development of scalable robot intelligence that captures complex behaviours efficiently for robust operation. Some research questions of interest that I wish to address are as follows

- RQ1:** How can geometric invariances be encoded in task representations to enable manipulation policies to generalise across variations in object geometry?
- RQ2:** How can physics-informed priors be incorporated into active manipulation policies to learn physically consistent and sample-efficient interactions in environments?
- RQ3:** How can spatiotemporal action-chunking encode inter-arm coordination while enabling policies to adapt to changing contact dynamics during bimanual manipulation?



**Early Research:** My early research at the Indian Institute of Science (IISc) focused on translating complex human intent into reliable robot actions through multimodal inputs for human-aware robot navigation. I developed a patented [2] bi-modal framework Brain Robot Interface, for audio-aware robot navigation, addressing the inherent stochasticity of biological signals. Simultaneously, I designed an inter-frame temporal module that recovered over 42% of missing human pose landmarks undetected by state-of-the-art estimators in varying environmental conditions (AIR 2023 [3]). I also contributed to a distributed denoising CNN for art restoration (ICETCI 2023 [4]) by constructing datasets that captured diverse degradation patterns.

Across these projects, I repeatedly encountered a common limitation: **models struggle to generalise when exposed to unseen variability** without the structural knowledge. This realisation motivated me to explore principled approaches for handling uncertainty and variability in robotics.

**Current Research in TU Delft:** I began formally exploring frameworks for uncertainty-aware and structure-informed learning. With Prof Javier Alonso-Mora's research group, I investigate **whole-body control of aerial manipulators** by deriving coupled Euler-Lagrange kinodynamics within a task-space NMPC framework. Unlike decoupled methods, this approach leverages the whole body dynamic structure for high-dexterity maneuvers. Furthermore, my work on disturbance-robust MPC utilised ellipsoidal terminal sets to provide formal asymptotic stability guarantees for underactuated systems under bounded perturbations. This research culminated in the **Best Master's Honours project** in the Faculty of Mechanical Engineering.

In addition to the research, coursework, especially on Lagrangian Neural Networks (LNN) and Gaussian Processes (GPs), reinforced a central insight shaping my research vision: **physically structured models are fundamental for reliable control**. While physical conservation laws and probabilistic priors act as powerful inductive biases for data efficiency and interpretability, their performance degrades under epistemic uncertainty in real-world environments. This tension between structural rigour and real-world uncertainty motivates my research direction: developing learning-enabled control frameworks that preserve underlying structure while explicitly adapting for uncertainty and environmental variability.

Under the guidance of Prof Jens Kober, I developed **CLIC-Chunks** [5], a framework that extends the **geometric policy shaping of Contrastive Policy Learning from Interactive Corrections** (CLIC) [6] to predict temporally coherent action chunks. The core of this approach is the construction of a **desired action chunk space**, which relaxes the traditional assumption of optimal human feedback by partitioning the action space into mutually exclusive desired and undesired regions. By centring the hypersphere at the human correction, with radii scaled proportionally to the robot-teacher distance, this geometric formulation treats the target as a continuous region rather than a rigid point, enabling robust exploration and preventing the policy from overfitting to suboptimal human feedback. Empirical results demonstrate that this approach can learn shorter action chunks while maintaining a high success rate. Our analysis also identifies a critical scalability bottleneck, revealing that Langevin MCMC sampling efficacy degrades when the total action chunk dimensionality increases.

Building on these findings, my Master's Thesis investigates **reward learning from suboptimal human corrections using geometrically structured action spaces**. I hypothesise that framing human interventions as positive Advantage bounds continuously across these geometrically structured action spaces yields a denser supervision signal than traditional preference-based learning. Theoretically grounded in potential-based reward shaping, this work aims to improve data efficiency by reducing the human intervention frequency compared to standard interactive imitation learning and also providing theoretical guarantees.

Concurrently, I extended Set-Supervised Diffusion Policy [7] towards **bimanual deformable object manipulation** for the WBCD Challenge at ICRA 2026. The framework learns action-chunking diffusion policies from contrastive correction pairs and is being adapted to dual-arm T-shirt manipulation tasks, strengthening my interest in robust deformable manipulation tasks. This work culminated the second position at the competition at Vienna.

My upcoming research will investigate **diffusion geometry** [8] to enable data-driven Riemannian calculus on noisy and low-density data, with applications to robotic manipulation. This research will be carried out under the supervision of Dr Sylvain Calinon at the Idiap Research Institute.

**Teaching, Outreaches & Leadership:** My experience in teaching and mentorship has been a primary vehicle for developing the **independence and collaborative mindset** essential for doctoral research. As a Teaching Assistant for the Intelligent Control Systems course by Prof Cosimo Della Santina, I supported students in implementing LNN, Iterative Learning Control, and GPs using JAX and PyTorch, sharpening my ability to translate complex theory into practice. Additionally, my involvement in hands-on robotics at IISc, workshops, and conference volunteering enhanced my ability to troubleshoot real-world systems and communicate complex ideas clearly. Together, these experiences have improved my technical versatility and prepared me to develop and validate research ideas in a PhD setting effectively.

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